

PITLYAR, Russian Federation

WMO: 234310

Lat: 65.8414N

Lon: 65.9125E

Elev: 17

StdP: 101.13

Time Zone: 5.00 (E05)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-42.8	-40.4	-45.2	0.0	-42.3	-42.7	0.1	-40.1	11.6	-12.5	10.0	-8.7	1.4	90	0.768

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.9	27.3	18.8	24.6	17.8	22.3	16.5	19.8	25.1	18.6	23.6	17.3	21.6	3.1	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.0	12.9	22.8	16.5	11.8	20.9	15.3	10.8	19.7	56.9	25.2	52.7	23.6	48.7	21.7	23.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.4	9.1	7.7	DB	-44.1	28.8	2.1	3.0	-45.7	30.9	-46.9	32.6	-48.1	34.3	-49.6	36.4	(4)
			WB	-43.9	20.9	1.8	1.6	-45.2	22.1	-46.2	23.1	-47.2	24.0	-48.5	25.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-4.1	-23.7	-20.9	-11.2	-6.4	1.5	11.9	16.6	12.1	6.3	-2.5	-14.6	-19.1
	DBStd	15.02	9.20	9.39	7.96	6.40	5.12	5.23	4.43	3.48	3.72	5.09	8.06	9.80
	HDD10.0	5533	1045	864	656	491	269	39	3	17	121	387	737	903
	HDD18.3	8215	1303	1098	914	741	521	200	88	194	362	646	987	1162
	CDD10.0	405	0	0	0	0	7	97	209	83	9	0	0	0
	CDD18.3	45	0	0	0	0	0	8	36	2	0	0	0	0
	CDH23.3	358	0	0	0	0	0	51	301	7	0	0	0	0
	CDH26.7	85	0	0	0	0	0	6	79	0	0	0	0	0
Wind	WSAvg	3.3	2.6	2.6	3.5	3.6	3.9	3.9	3.3	3.4	3.5	3.6	2.8	2.7
Precipitation	PrecAvg	875	56	49	68	70	72	79	79	86	72	105	75	64
	PrecMax	1036	88	80	148	144	115	116	161	171	144	150	129	97
	PrecMin	758	37	19	20	29	31	34	16	37	22	39	48	39
	PrecStd	62	13	16	21	23	22	23	32	31	26	23	19	16
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-1.7	-1.0	4.1	7.3	19.6	27.4	31.1	24.1	18.7	9.7	1.8	-0.3
		MCWB	-2.4	-1.8	1.1	3.6	13.5	18.3	20.0	17.8	13.9	7.5	1.0	-1.2
	2%	DB	-4.8	-3.5	2.1	5.7	16.1	24.6	29.0	21.6	15.4	7.2	-0.5	-2.5
		MCWB	-5.4	-4.2	0.0	2.6	10.5	16.4	19.6	16.0	11.8	5.8	-1.3	-3.2
	5%	DB	-7.7	-5.6	0.8	4.0	12.2	22.5	26.9	20.0	13.3	5.2	-1.9	-4.2
		MCWB	-8.2	-6.2	-0.8	1.8	8.1	15.5	19.0	15.1	10.8	3.9	-2.5	-4.7
	10%	DB	-10.5	-8.0	-0.5	2.5	8.8	20.1	24.5	18.4	11.5	3.9	-4.1	-6.1
		MCWB	-11.0	-8.6	-1.7	0.7	5.6	14.6	18.2	14.5	9.7	3.0	-4.6	-6.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-2.3	-1.6	1.7	4.6	13.6	19.6	22.4	18.9	14.5	8.1	1.0	-0.8
		MCDB	-1.8	-1.0	3.7	5.9	18.6	24.9	28.0	23.1	18.3	8.7	1.9	-0.4
	2%	WB	-5.5	-4.2	0.3	2.9	10.8	17.9	20.8	17.3	12.5	5.7	-1.2	-3.1
		MCDB	-5.0	-3.7	1.7	5.2	15.8	22.5	25.6	20.1	14.6	7.2	-0.7	-2.5
	5%	WB	-8.3	-6.1	-0.7	1.8	8.4	16.5	19.6	16.1	11.2	4.1	-2.5	-4.6
		MCDB	-7.8	-5.5	0.6	3.8	12.3	21.3	25.0	18.9	12.8	5.0	-1.9	-4.2
	10%	WB	-10.9	-8.5	-1.7	0.8	5.7	15.0	18.5	15.0	9.8	2.9	-4.6	-6.4
		MCDB	-10.3	-8.0	-0.4	2.4	8.3	19.9	23.8	17.9	11.3	3.7	-4.1	-6.1
Mean Daily Temperature Range	5% DB	MDBR	8.0	8.1	9.2	9.3	7.2	8.2	8.9	7.6	5.8	4.7	7.3	7.6
		MCDBR	9.7	7.5	6.8	8.2	11.5	10.9	12.3	10.4	8.6	5.7	7.5	6.5
		MCWBR	9.5	7.1	5.4	6.0	6.9	5.6	5.0	6.1	5.6	4.3	6.9	6.3
	5% WB	MCDBR	9.6	7.6	6.9	7.8	11.0	9.9	10.8	8.7	7.6	4.9	7.3	6.6
		MCWBR	9.4	7.2	5.7	5.9	7.0	5.9	5.2	6.0	5.3	4.1	6.9	6.4
Clear-Sky Solar Irradiance	taub	0.207	0.264	0.312	0.336	0.353	0.369	0.408	0.376	0.325	0.285	0.204	0.214	
	taud	1.886	2.043	2.047	2.099	2.276	2.431	2.310	2.373	2.482	2.255	1.925	1.898	
	Ebn at Noon	223	582	727	815	846	842	787	778	749	569	217	26	
	Edh at Noon	20	59	97	119	111	99	108	92	66	48	18	5	
All-Sky Solar Radiation	RadAvg	0.09	0.69	2.24	4.24	5.36	5.18	5.21	3.38	1.84	0.68	0.16	0.02	
	RadStd	0.01	0.07	0.15	0.26	0.36	0.41	0.55	0.28	0.18	0.09	0.02	0.00	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page